

# RITHVIK REDDY VELAPATI

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## Summary

Full-Stack Developer with nearly four years of experience building enterprise, web, mobile, and AI-powered applications. Strong experience in Java, Spring Boot, Node.js, NestJS, React, Next.js, REST APIs, microservices, PostgreSQL, MongoDB, AWS, and Azure. Progressed from Full-Stack Software Engineer to Lead Developer at Healthvive, leading backend, frontend, mobile, AI, and cloud delivery. Experienced in OAuth 2.0 authentication, event-driven systems, Docker, Kubernetes, and CI/CD.

## Technical Skills

**Languages:** Java, JavaScript, TypeScript, Python, SQL  
**Frontend:** React.js, Next.js, Angular, React Native, HTML5, CSS3, Tailwind CSS  
**Backend:** Node.js, NestJS, Spring Boot, Spring MVC, Bun, REST APIs, GraphQL, Microservices  
**Frameworks & ORM:** Spring Data JPA, Hibernate, TypeORM  
**Databases & Caching:** MongoDB, PostgreSQL, MySQL, SQL Server, Oracle SQL, Azure SQL, Azure Cosmos DB, Redis  
**AI & LLM:** OpenAI API, Amazon Bedrock, Titan Embeddings, Texttract, OpenSearch, Replicate, RAG, Zod, Instructor, Structured Outputs, LLM Evaluations

**Cloud & DevOps:** Azure, Azure DevOps, AKS, ACR, Blob Storage, Terraform, AWS EC2, S3, RDS, Lambda, API Gateway, CloudWatch, Docker, Kubernetes, Jenkins, CI/CD  
**Authentication & Security:** Firebase Authentication, OAuth 2.0, JWT, RBAC, OWASP  
**Messaging & Integration:** Apache Kafka, SQS, SNS, Webhooks, Socket.IO  
**Mobile:** Capacitor, Android Studio, Xcode, Firebase Cloud Messaging  
**Testing & Tools:** JUnit, Mockito, Jest, Postman, Swagger/OpenAPI, Sentry, Maven, Git, GitHub  
**Collaboration:** Agile, Scrum, Kanban, Jira, Confluence, Figma

## Experience

### Lead Developer (Promoted from Full-Stack Software Engineer) Healthvive Inc. (WellFed App)

**08/2024 - Present**  
**Woodstock, Georgia**

- Architected and lead WellFed from the ground up, delivering an AI-powered nutrition and social meal-planning platform across web, iOS, and Android while leading backend, frontend, mobile, AI, and DevOps delivery.
- Design and develop backend services using Node.js, Java, TypeScript, NestJS, and Bun, and build REST and GraphQL APIs for recipes, users, collections, subscriptions, meal planning, and social features.
- Implement Firebase Authentication, OAuth 2.0 social login, JWT and ID-token verification, RBAC, secure cookies, request validation, and server-to-server authentication.
- Apply OWASP security practices through API rate limiting, CORS configuration, input sanitization, secure HTTP headers, secrets management, and webhook signature verification.
- Develop event-driven and real-time workflows using Kafka, webhooks, and Socket.IO for AI processing, notifications, nutrition data, subscription events, and asynchronous operations.
- Design MongoDB schemas and TypeORM entities, relationships, indexes, and query patterns and implement Redis caching to reduce repeated database queries and third-party API calls.
- Integrate NestJS services with React and Next.js and build responsive interfaces using TypeScript, Tailwind CSS, and WCAG accessibility practices.
- Develop personalized recipe and nutrition workflows using OpenAI API, Replicate, prompt engineering, Zod, Instructor, structured outputs, and LLM evaluations.
- Build production RAG pipelines using embeddings, vector search, guardrails, prompt evaluation, and monitoring to deliver relevant and reliable AI-generated responses.
- Provision and manage cloud infrastructure using Terraform, including Kubernetes, networking, storage, secrets, and environment-specific configurations.
- Deploy and operate cloud services across AWS using EC2, S3, RDS, Lambda, API Gateway, and CloudWatch, configuring scalable compute, storage, databases, serverless workflows, API integrations, monitoring, and environment-specific deployments.
- Deploy Dockerized services using AKS, ACR, Blob Storage, and Azure DevOps CI/CD; write tests using Jest and monitor production systems using Sentry and application logs.

### Software Developer Unique Logic Solutions

**05/2024 - 08/2024**  
**Bentonville, Arkansas**

- Developed an online pharmacy platform using Java, Spring Boot, React, REST APIs, PostgreSQL, and AWS for prescription submission, pharmacist review, medication ordering, and payment workflows.

- Built Spring Boot microservices using Spring MVC with controller, service, and repository layers for prescription, medication, order, and payment modules.
- Created Spring Data JPA repositories and Hibernate entity mappings for transactional healthcare data using PostgreSQL and MySQL.
- Optimized SQL queries, joins, indexes, and database-access patterns for frequently accessed prescription, medication, customer, and order records.
- Developed Kafka producers and consumers for asynchronous prescription reviews, medication orders, payment processing, and application events.
- Built reusable React components using HTML5, CSS3, JavaScript, and TypeScript and integrated them with Spring Boot REST APIs.
- Implemented frontend form validation, API data mapping, loading states, and standardized error handling for pharmacy workflows.
- Developed a Python-based AI-assisted pharmacist-review workflow using AWS Lambda, Amazon S3, Textract, Titan Embeddings, OpenSearch, and Amazon Bedrock.
- Deployed containerized application services using Docker, AWS EC2, S3, and RDS and monitored application logs, errors, and performance through CloudWatch.
- Wrote unit and integration tests using JUnit and Mockito and used Postman and Swagger/OpenAPI to test and document REST APIs.
- Maintained Maven and Jenkins CI/CD pipelines and used Git, Jira, and Confluence for builds, deployments, code reviews, defect tracking, and Agile delivery.

**Java Software Developer**  
**Cognizant**

**01/2021 - 07/2022**  
**Hyderabad, India**

- Developed and maintained an internal employee-management application using Java, Spring Boot, Node.js, React, REST APIs, and PostgreSQL for onboarding, employee records, and performance-evaluation workflows.
- Built backend components using Java, Spring Boot, and Spring MVC with controller, service, and repository layers to maintain separation of concerns.
- Developed REST endpoints for HR modules, implementing request validation, business rules, response mapping, centralized exception handling, and consistent API contracts.
- Developed Node.js services for notifications, document processing, scheduled tasks, and asynchronous application workflows.
- Created Hibernate and Spring Data JPA entity mappings, repository queries, and transaction workflows using PostgreSQL and MySQL.
- Designed PostgreSQL tables, entity relationships, joins, indexes, and optimized queries for employee and performance-management data.
- Used MongoDB collections for modules requiring flexible document structures and schema-independent data storage.
- Developed reusable frontend components and forms using React, Angular, HTML5, CSS3, JavaScript, and integrated them with Java and Node.js REST APIs.
- Deployed Java and Node.js application services using AWS EC2, stored employee documents in Amazon S3, and managed PostgreSQL databases using Amazon RDS.
- Used AWS Lambda and API Gateway for serverless processing and API access and implemented asynchronous notifications and background operations using Amazon SQS and SNS.
- Monitored application logs, errors, and infrastructure metrics using Amazon CloudWatch and containerized application services using Docker.
- Wrote tests using JUnit, Mockito, and Jest; maintained Maven and Jenkins CI/CD pipelines; and used Git, Jira, and Confluence for code reviews, deployments, defect tracking, and Agile delivery.

Academic Projects

**AI-Enhanced Vintage Car Auction Platform:** Built a full-stack platform using Java, Spring Boot, React, Python, PostgreSQL, Amazon Bedrock, Titan Embeddings, and OpenSearch. It supports vehicle listings, auction scheduling, bid tracking, semantic search, and personalized recommendations.

**Mobile Ride and Food-Booking Application:** Built a cross-platform application using React Native, Node.js, Python, REST APIs, MySQL, and MongoDB. It supports ride booking, restaurant browsing, food ordering, payments, location analysis, and order tracking.

Education

**Master of Science: Computer Science**  
University of Memphis

**Bachelor of Technology: Computer Science**  
Anurag University

**05/2024**  
Memphis, TN

**04/2022**  
Hyderabad, India

Certifications

- AWS Certified Developer - Associate
- Microsoft Engineering Virtual Experience Program — Forage, May 2021

- Complete Java Full-Stack Developer and Complete Web Developer — Udemy
- Applied Machine Learning in Python — University of Michigan, Coursera